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Exact Erasure Tails and Delay Bounds for Store-Carry-Forward Emergency Messaging

Binomial and Poisson-binomial tails, exponential delay, and conservative bounds

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Abstract

Offline emergency messaging combines two uncertainties: a message split into shards is reconstructible only if enough shards arrive, and a store-carry-forward relay delivers only when a contact occurs. This report derives the exact reconstruction probability for independent and heterogeneous shard success, the exact delay distribution for independent relay copies under Poisson contacts, and conservative Chernoff and Hoeffding bounds for the erasure tail. We show where the bounds are useful and where they overprovision. Numerical tables are generated from the SENDA reference implementation and are model outputs, not field measurements. The report is a technical reference for the offline stack; it does not claim that the independence or exponential-contact assumptions hold in a particular deployment.

Keywords: erasure coding, Poisson-binomial, delay-tolerant networking, Chernoff bound, Hoeffding bound, offline messaging.

1. Introduction

A store-carry-forward system does not have an end-to-end path. It has contacts, copies, storage, and time [1, 2, 3]. Two design questions dominate:

1. **Reconstruction.** If a message is encoded into n shards and any k shards suffice [4, 5], what is the probability that at least k shards arrive?
2. **Deadline.** If several independent copies circulate and contacts are Poisson, what is the probability that at least one copy is delivered by a deadline t?

The first question is a binomial tail when shard successes are iid, and a Poisson-binomial tail when they are not [6]. The second is the distribution of the minimum of independent exponential contact times. Both have closed forms. The engineering temptation is to replace them with a simple bound; that is sometimes right and sometimes expensive.

This report makes the assumptions explicit, derives the formulas used by the SENDA precision lab, and quantifies the gap between exact tails and two standard bounds. Section 2 states the model. Section 3 derives erasure tails. Section 4 derives delay. Section 5 derives bounds. Section 6 reports numerical tables. Section 7 gives guidance. Section 8 lists limitations.

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2. Model and assumptions

We use the following assumptions.

- **A1 (Shard independence).** Shard delivery indicators $X_1, \dots, X_n$ are independent Bernoulli random variables.
- **A2 (Homogeneity, optional).** In the iid case, $\Pr(X_i = 1) = p$ for all i. In the heterogeneous case, $\Pr(X_i = 1) = p_i$.
- **A3 (Poisson contacts).** For each relay copy, contact times form a Poisson process with rate $\lambda > 0$; copies are independent.
- **A4 (No correlation, no recovery).** A failed shard is not retransmitted within the modeled window, and contact events do not correlate across copies.

A1 and A4 are strong. Real outages cluster, and copies may share a vehicle or a power source. The formulas below are therefore a baseline, not a field prediction. A deployment that observes correlation should replace p_i with a correlated model or report the gap as a sensitivity.

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3. Exact erasure tails

3.1 Independent and identically distributed shards

Let $K = \sum_{i=1}^{n} X_i$ with $X_i \sim \mathrm{Bernoulli}(p)$ independent. Then $K \sim \mathrm{Binomial}(n, p)$ and

$$\Pr(K \geq k) = \sum_{j=k}^{n} \binom{n}{j} p^{j} (1-p)^{n-j}. \tag{1}$$

Theorem 1. Under A1 and A2, the reconstruction probability of a k-of-n erasure code is given by (1).

The proof is the definition of the binomial distribution. The formula is exact in real arithmetic; the implementation evaluates the terms in log space and clamps the result to $[0, 1]$ to control floating-point error.

3.2 Heterogeneous shards

When shard success probabilities differ, K is Poisson-binomial:

$$\Pr(K = j) = \sum_{\substack{S \subseteq \{1,\dots,n\} \\ |S| = j}} \prod_{i \in S} p_i \prod_{i \notin S} (1 - p_i). \tag{2}$$

Direct enumeration is exponential. The standard dynamic program is $O(n^2)$.

Theorem 2. Let $f_i(j) = \Pr(\sum_{m=1}^{i} X_m = j)$ with $f_0(0) = 1$ and $f_0(j) = 0$ for $j \neq 0$. Then

$$f_{i+1}(j) = f_i(j)(1 - p_{i+1}) + f_i(j-1) p_{i+1}, \tag{3}$$

and $\Pr(K \geq k) = \sum_{j=k}^{n} f_n(j)$.

Proof. Condition on X_{i+1}. If it fails, the first i variables must sum to j; if it succeeds, they must sum to $j-1$. The events are disjoint and exhaustive. ∎

The SENDA implementation stores the distribution as a length-$i+1$ array and updates it in place per shard. For $n \leq 255$ the memory is negligible.

3.3 Numerical notes

- For small p and large n, direct powers underflow. The iid implementation computes each term as $\exp(\log \Gamma(n+1) - \log \Gamma(j+1) - \log \Gamma(n-j+1) + j \log p + (n-j)\log(1-p))$.
- The heterogeneous DP avoids underflow longer because it multiplies probabilities rather than exponentiating large log-factorials, but it can still underflow for very small tail probabilities. A log-domain DP is the correct next step if tails below 10^{-300} matter.
- Clamping to $[0, 1]$ hides a negative round-off but does not fix a bad model. A production verifier should report the unclamped value and a tolerance.

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4. Delay under Poisson contacts

4.1 One copy

Let T_1 be the first contact time of a copy with rate λ. Then $\Pr(T_1 > t) = e^{-\lambda t}$, so

$$F_{T_1}(t) = 1 - e^{-\lambda t}, \qquad \mathbb{E}[T_1] = \frac{1}{\lambda}. \tag{4}$$

4.2 Independent copies

Let $T = \min(T_1, \dots, T_m)$ for m independent copies with the same rate λ. The minimum is exponential with rate $m\lambda$:

$$\Pr(T > t) = \prod_{i=1}^{m} \Pr(T_i > t) = e^{-m\lambda t}. \tag{5}$$

Theorem 3. Under A3, with m independent copies of rate λ,

$$F_T(t) = 1 - e^{-m\lambda t}, \qquad \mathbb{E}[T] = \frac{1}{m\lambda}, \qquad q_\alpha = \frac{-\ln(1-\alpha)}{m\lambda}. \tag{6}$$

Proof. The survival product is (5). The CDF is its complement. The mean and quantiles follow from the exponential distribution. ■

The reference implementation uses (6) directly. It reports the median ($\alpha = 0.5$), the 90th percentile, and the 99th percentile.

4.3 Per-contact delivery probability

If a contact occurs but the transfer succeeds only with probability q, the effective contact process is thinned to rate λq. Replacing λ by λq in (6) gives

$$F_T(t) = 1 - e^{-m\lambda q t}. \tag{7}$$

Equation (7) is the model to use when contacts are noisy. The current reference implementation exposes the $q = 1$ case; the extension is a substitution, not a new derivation.

4.4 Combined erasure and delay

If reconstruction requires k of n shards and each shard has an independent deadline-t delivery probability $p(t) = 1 - e^{-\lambda q t}$ (one copy per shard), then the reconstruction probability by t is the binomial tail (1) with $p = p(t)$. With m copies per shard, use $p(t) = 1 - e^{-m\lambda q t}$. This composition assumes independence between shards and copies; it is the natural baseline, not a theorem about a real network.

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5. Conservative bounds

5.1 Chernoff bound

For $X \sim \mathrm{Binomial}(n, p)$ and $a = k/n > p$, the Chernoff bound [7] is

$$\Pr(X \geq k) \leq \exp\left(-n\, D(a \| p)\right), \qquad D(a\|p) = a \ln \frac{a}{p} + (1-a) \ln \frac{1-a}{1-p}. \tag{8}$$

Proof sketch. For $t > 0$, Markov's inequality gives $\Pr(X \geq k) \leq e^{-tk}(1 - p + pe^t)^n$. Minimizing the exponent over t yields $e^t = a(1-p)/(p(1-a))$ and the KL divergence $D(a\|p)$. ∎

For $a < p$, the same bound applies to the lower tail with $D(a\|p)$.

5.2 Hoeffding bound

Hoeffding's inequality [8] gives, for $\epsilon > 0$,

$$\Pr(X \geq n(p + \epsilon)) \leq e^{-2n\epsilon^2}. \tag{9}$$

With $a = k/n$, set $\epsilon = a - p$. The bound is looser than Chernoff because it uses only the range of the Bernoulli variable, not its variance.

5.3 When to use which

- Use the exact tail when $n \leq 255$ and the independence assumption is defensible. It is cheap and avoids unnecessary redundancy.
- Use Chernoff for a quick conservative design when n is large or when only a bound is needed.

- Use Hoeffding when its simpler form is required for an analytical argument, and accept the extra conservatism.
- Never present a bound as an exact probability. The display contract should say `≤` or `<`, not `=`.

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6. Numerical study

All values below are computed by the SENDA reference implementation. They are model outputs under A1–A4, not measurements. Tables are reproducible with `node scripts/research/paper2-tables.mjs`.

6.1 Homogeneous erasure tail

$n = 32$, $p = 0.2$. The exact tail is compared with Chernoff (8) and Hoeffding (9).

K	EXACT $\mathrm{PR}(K \geq K)$	CHERNOFF BOUND	HOEFFDING BOUND
8	0.301763	0.789604	0.852144
10	0.0898324	0.323430	0.444858
12	0.0167163	0.0738217	0.140858
14	0.00195246	0.00986601	0.0270518
16	0.000143595	0.000792282	0.00315111
20	1.89197×10^{-7}	1.12643×10^{-6}	9.54016×10^{-6}

The bounds are valid but increasingly conservative. At $k = 12$ the Chernoff bound is about 4.4 times the exact tail; Hoeffding is about 8.4 times. Designing to the bound overprovisions redundancy.

6.2 Heterogeneous erasure tail

$n = 32$, $p_i = 0.15 + 0.01(i - 1)$ for $i = 1, \ldots, 32$, so the probabilities range from 0.15 to 0.46. The exact Poisson-binomial tail is computed by (3).

K	EXACT $\mathrm{PR}(K \geq K)$
10	0.531164
12	0.243764
14	0.0740366
16	0.0144282
18	0.00176455
20	0.000132207

A homogeneous model with the mean probability $\bar{p} = 0.305$ would not reproduce this tail; heterogeneity matters most in the upper tail.

6.3 Delay by deadline

$\lambda = 0.2$ contacts per hour, independent copies, $q = 1$. The probability column is $1 - e^{-m\lambda t}$; the percentiles are in hours.

COPIES M	DEADLINE T (H)	$\mathrm{PR}(T \leq T)$	MEAN	MEDIAN	P90	P99
1	6	0.698806	5.000	3.466	11.513	23.026
1	12	0.909282	5.000	3.466	11.513	23.026
1	24	0.991770	5.000	3.466	11.513	23.026
2	6	0.909282	2.500	1.733	5.756	11.513
2	12	0.991770	2.500	1.733	5.756	11.513
3	6	0.972676	1.667	1.155	3.838	7.675
3	12	0.999253	1.667	1.155	3.838	7.675
5	6	0.997521	1.000	0.693	2.303	4.605
5	12	0.999994	1.000	0.693	2.303	4.605

Doubling copies halves the expected delay but does not make the tail free: the 99th percentile also halves. The cost is linear in copies.

6.4 Erasure overhead

k-of-n with overhead $n/k = 1.5$.

K	N	P	$\mathrm{PR}(K \geq K)$	EXPECTED RECEIVED
8	12	0.4	0.0573099	4.8
8	12	0.6	0.438178	7.2
8	12	0.8	0.927445	9.6
12	18	0.6	0.374277	10.8
16	24	0.6	0.327922	14.4

At $p = 0.6$ and 50% overhead, the reconstruction probability is below one half. The expected number of received shards is $7.2 < 8$, which is a useful warning: a mean above the threshold is not enough, because the threshold is a tail event.

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7. Engineering guidance

1. **Separate the model from the claim.** Publish n, k, p_i, λ, m, and q as inputs. Publish the tail as an exact value or a bound with an explicit inequality.
2. **Prefer exact tails in the supported range.** For $n \leq 255$ the DP or log-sum is cheap. A bound is a design aid, not a substitute for the exact value.
3. **Report heterogeneity.** A single mean p can hide a much heavier tail when shard probabilities vary.
4. **Do not confuse expected received shards with success probability.** The threshold is a tail, not a mean.
5. **Treat Poisson contacts as a baseline.** If contacts cluster, the exponential model underestimates long delays.
6. **State the deadline and the copies in the display.** “99% by 12 hours with 3 copies” is checkable; “reliable” is not.

8. Limitations

- Independence (A1) and exponential contacts (A3) are modeling assumptions.
- The reference implementation uses double-precision arithmetic, not exact rational arithmetic. The formulas are exact; the computed values are not.
- The $O(n^2)$ Poisson-binomial DP is appropriate for the supported $n \leq 255$; larger systems need a faster or log-domain method.
- Correlation, retransmission, storage limits, and adversarial drop are outside the model.
- The numerical tables are not field data and must not be cited as operational evidence.

9. Conclusion

The exact erasure tail and the exponential delay distribution are simple enough to implement and verify offline. The main risk is not the formula; it is silently replacing it with a mean or a loose bound and then treating the result as a promise. PCI-style claims make the inputs, the method, the inequality, and the provenance explicit, so a reviewer can see which assumption is doing the work.

References

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Appendix A — Reproduction

```
node scripts/research/paper2-tables.mjs
```

The script imports the same functions used by the precision lab, writes a JSON artifact under `public/research/data/`, and prints the tables above. It performs no network access.